

Prove existence of a simpler all-norms approximation clustering for correlation clustering

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Website #27 | Solver verdict: SOLVED | Verifier verdict: ISSUES | Shown: solver output

Problem

We are given a complete graph on a finite vertex set V , each edge labeled $+$ or $-$. For a clustering $\mathcal{C}$ (a partition of V), the disagreement degree of a vertex u is

$$y_{\mathcal{C}}(u) = |\{v : uv \in E^+, \mathcal{C}(u) \neq \mathcal{C}(v)\}| + |\{v : uv \in E^-, \mathcal{C}(u) = \mathcal{C}(v)\}|.$$

Let $y_{\mathcal{C}} \in \mathbb{Z}_{\geq 0}^V$ be the disagreement vector. For $p \in [1, \infty]$,

$$\text{OPT}_p := \min_{\mathcal{C}} \|y_{\mathcal{C}}\|_p.$$

The task is to prove existence of a *single* clustering $\hat{\mathcal{C}}$ and a universal constant α such that

$$\|y_{\hat{\mathcal{C}}}\|_p \leq \alpha \text{OPT}_p \quad \text{for every } p \in [1, \infty].$$

I will in fact prove a stronger statement: one clustering simultaneously 65-approximates every top- k objective, hence every ℓ_p -objective.

Alternative readings considered

The statement is unambiguous. The only minor bookkeeping choice is whether to allow positive self-loops in the auxiliary graph used in the proof; I will add them purely for analysis. They never contribute to disagreement costs and hence do not alter the original optimization problem.

Related literature found

- **Davies–Moseley–Newman (ICALP 2024)** — SOLVES. They prove the exact statement asked here, algorithmically, via an adjusted correlation metric and a long ℓ_p -specific charging analysis; in their conclusion they explicitly ask whether there is a simpler existential proof. (drops.dagstuhl.de)
- **Cao–Li–Ye (ICALP 2025)** — SOLVES a stronger statement (all monotone symmetric norms) using a much simpler top- k -based construction and a cleaner decomposition. The proof below is guided by that top- k viewpoint, but I re-derive the needed inequalities directly here and import only the rounding theorem. (drops.dagstuhl.de)
- **Kalhan–Makarychev–Zhou (NeurIPS 2019)** — TOOL. Their metric-rounding algorithm on complete graphs yields a vertex-by-vertex 5-factor bound from any feasible metric LP solution; this is the only external theorem used inside the proof. (papers.neurips.cc)
- **Puleo–Milenkovic (arXiv 2016)** — BACKGROUND/TOOL. Earlier local-error rounding for complete graphs; useful context, but the sharper 5-factor per-vertex rounding above makes the present argument cleaner. (arxiv.org)

Proof sketch

The clean way to make one solution good for all p is to work with the *top- k* seminorms

$$\text{Top}_k(z) := \sum_{i=1}^k z_i^\downarrow,$$

where $z^\downarrow$ is the nonincreasing rearrangement. If a single vector is within factor ρ of optimum for every top- k , then it is within factor ρ for every ℓ_p as well; I give a short direct proof of this from duality, so no majorization theorem is needed.

Thus it suffices to construct one fractional metric x whose fractional disagreement vector has constant-factor approximation for every top- k , and then round x to an integral clustering with only constant *per-vertex* loss. The rounding step is exactly where the Kalhan–Makarychev–Zhou theorem enters.

The fractional metric is simple. Write $N(u)$ for the positive neighborhood of u (with u itself added as a positive self-loop for bookkeeping), and let $d(u) = |N(u)|$. Remove a positive edge uv if the positive neighborhoods of u and v differ too much:

$$|N(u) \Delta N(v)| > \beta \max\{d(u), d(v)\},$$

with $\beta = \frac{1}{2}$. Let H be the surviving positive graph. Then define

$$x_{uv} = 1 - \frac{|N_H(u) \cap N_H(v)|}{\max\{d(u), d(v)\}}.$$

This says: two vertices are close if, after discarding badly mismatched positive edges, they still share many positive neighbors.

For a fixed top- k -optimal clustering $\mathcal{C}$, and U the k vertices with largest fractional costs under x , I decompose the contributions to $\sum_{u \in U} \text{cost}_x(u)$ into five edge types relative to $\mathcal{C}$: true positive disagreements, true negative disagreements, surviving positive agreements, discarded positive agreements, and cut negative agreements. The true disagreements are easy. The discarded positive agreements are charged directly to local disagreements using the large neighborhood-symmetric-difference condition. The surviving positive agreements are charged partly to actual disagreements and partly to discarded positive agreements. The cut negative edges are charged through common neighbors in H . In each case, the charge has coefficient vector c with both $\|c\|_\infty$ and $\|c\|_1/k$ bounded by a universal constant; a simple convex-combination lemma then converts this into a top- k bound.

Choosing $\beta = \frac{1}{2}$ gives fractional top- k cost at most 13 times optimum. The external rounding theorem loses another factor 5, so the final clustering is a simultaneous 65-approximation for all top- k , hence for all ℓ_p .

Main result

Theorem 1. *For every unweighted complete signed graph $G = (V, E^+ \cup E^-)$, there exists a clustering $\hat{\mathcal{C}}$ such that for every $k \in \{1, \dots, n\}$,*

$$\text{Top}_k(y_{\hat{\mathcal{C}}}) \leq 65 \cdot \text{OPT}_k^{\text{top}},$$

where

$$\text{OPT}_k^{\text{top}} := \min_{\mathcal{C}} \text{Top}_k(y_{\mathcal{C}}).$$

Consequently, for every $p \in [1, \infty]$,

$$\|y_{\hat{\mathcal{C}}}\|_p \leq 65 \text{OPT}_p.$$

Proof

For a nonnegative vector $z \in \mathbb{R}_{\geq 0}^V$, write $z_1^\downarrow \geq \dots \geq z_n^\downarrow$ for its coordinates in nonincreasing order and

$$\text{Top}_k(z) := \sum_{i=1}^k z_i^\downarrow = \max_{\substack{S \subseteq V \\ |S|=k}} \sum_{u \in S} z(u).$$

Lemma 1. *Let $a : V \rightarrow \mathbb{R}_{\geq 0}$, let $k \in \{1, \dots, n\}$, and suppose*

$$\|a\|_\infty \leq \alpha, \quad \|a\|_1 \leq \alpha k.$$

Then for every nonnegative vector $z \in \mathbb{R}_{\geq 0}^V$,

$$\sum_{u \in V} a(u)z(u) \leq \alpha \text{Top}_k(z).$$

Proof. Set $b = a/\alpha$. Then $0 \leq b(u) \leq 1$ for every u , and $\sum_u b(u) \leq k$. Hence b lies in the polytope

$$P_k := \{x \in [0, 1]^V : \sum_{u \in V} x(u) \leq k\}.$$

Every extreme point of P_k is a $\{0, 1\}$ -vector with at most k ones, so b is a convex combination of such vectors:

$$b = \sum_t \lambda_t \mathbf{1}_{S_t}, \quad \lambda_t \geq 0, \quad \sum_t \lambda_t = 1, \quad |S_t| \leq k.$$

Therefore

$$\sum_u a(u)z(u) = \alpha \sum_u b(u)z(u) = \alpha \sum_t \lambda_t \sum_{u \in S_t} z(u) \leq \alpha \sum_t \lambda_t \text{Top}_k(z) = \alpha \text{Top}_k(z). \quad \square$$

We now build the fractional metric.

For bookkeeping only, add a positive self-loop uu at every vertex u . This changes nothing in the original problem: self-loops are never disagreement edges, and later we set $x_{uu} = 0$. Let

$$N(u) := \{u\} \cup \{v \in V \setminus \{u\} : uv \in E^+\}, \quad d(u) := |N(u)| \geq 1,$$

and for $u, v \in V$,

$$M_{uv} := \max\{d(u), d(v)\}.$$

Fix

$$\beta := \frac{1}{2}.$$

Define the auxiliary positive graph $H = (V, E_H)$ by

$$E_H := \{uv \in E^+ : |N(u) \triangle N(v)| \leq \beta M_{uv}\},$$

where all self-loops uu belong to E_H . Let $N_H(u)$ be the H -neighborhood of u . Define $x \in [0, 1]^{V \times V}$ by

$$x_{uu} := 0, \quad x_{uv} := 1 - \frac{|N_H(u) \cap N_H(v)|}{M_{uv}} \quad (u \neq v).$$

Lemma 2. *The array x is a metric on V : it is symmetric, satisfies $x_{uu} = 0$, lies in $[0, 1]$, and obeys the triangle inequality*

$$x_{uv} + x_{uw} \geq x_{vw} \quad \forall u, v, w \in V.$$

Proof. Symmetry and $x_{uu} = 0$ are immediate. Also

$$0 \leq |N_H(u) \cap N_H(v)| \leq \min\{d(u), d(v)\} \leq M_{uv},$$

so $0 \leq x_{uv} \leq 1$.

It remains to prove the triangle inequality. Assume u, v, w are distinct and, without loss of generality, $d(v) \geq d(w)$. Then

$$\begin{aligned} x_{uv} + x_{uw} - x_{vw} &= \left(1 - \frac{|N_H(u) \cap N_H(v)|}{M_{uv}}\right) + \left(1 - \frac{|N_H(u) \cap N_H(w)|}{M_{uw}}\right) \\ &\quad - \left(1 - \frac{|N_H(v) \cap N_H(w)|}{d(v)}\right) \\ &= 1 + \frac{|N_H(v) \cap N_H(w)|}{d(v)} - \frac{|N_H(u) \cap N_H(v)|}{M_{uv}} - \frac{|N_H(u) \cap N_H(w)|}{M_{uw}} \\ &\geq 1 + \frac{|N_H(v) \cap N_H(w)|}{M_{uv}} - \frac{|N_H(u) \cap N_H(v)|}{M_{uv}} - \frac{|N_H(u) \cap N_H(w)|}{M_{uw}} \\ &\geq 1 - \frac{|N_H(u) \setminus N_H(w)|}{M_{uv}} - \frac{|N_H(u) \cap N_H(w)|}{M_{uw}}. \end{aligned}$$

The second inequality uses

$$|N_H(u) \cap N_H(v)| - |N_H(v) \cap N_H(w)| \leq |N_H(u) \setminus N_H(w)|.$$

Since $M_{uv} \geq d(u)$ and $M_{uw} \geq d(u)$, we get

$$x_{uv} + x_{uw} - x_{vw} \geq 1 - \frac{|N_H(u) \setminus N_H(w)| + |N_H(u) \cap N_H(w)|}{d(u)} = 1 - \frac{|N_H(u)|}{d(u)} \geq 0,$$

because $N_H(u) \subseteq N(u)$, so $|N_H(u)| \leq d(u)$. This proves the triangle inequality. $\square$

For a metric x , define the fractional disagreement at u by

$$\text{cost}_x(u) := \sum_{v: uv \in E^+} x_{uv} + \sum_{v: uv \in E^-} (1 - x_{uv}).$$

(The self-loop contributes $x_{uu} = 0$, so it is harmless.)

Fix $k \in \{1, \dots, n\}$. Let

$$\text{Top}_k(x) := \max_{|U|=k} \sum_{u \in U} \text{cost}_x(u).$$

Choose $U \subseteq V$, $|U| = k$, attaining this maximum. Also fix a clustering $\mathcal{C}$, and later specialize $\mathcal{C}$ to a top- k -optimal clustering. For $u \in V$, write $\mathcal{C}(u)$ for the cluster of u , and write $\text{cost}_{\mathcal{C}}(u)$ for its disagreement degree under $\mathcal{C}$.

We partition all unordered pairs into five classes relative to $\mathcal{C}$:

$$\begin{aligned} \phi_1^+ &:= \{uv \in E^+ : \mathcal{C}(u) \neq \mathcal{C}(v)\}, \\ \phi_2^+ &:= \{uv \in E^+ \setminus E_H : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ \phi_3^+ &:= \{uv \in E_H : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ \phi_1^- &:= \{uv \in E^- : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ \phi_2^- &:= \{uv \in E^- : \mathcal{C}(u) \neq \mathcal{C}(v)\}. \end{aligned}$$

(Thus $\phi_1^+ \cup \phi_2^+ \cup \phi_3^+ = E^+$, and $\phi_1^- \cup \phi_2^- = E^-$.)

For $u \in V$, let $\phi_i^\pm(u)$ be the set of pairs in $\phi_i^\pm$ incident to u , and let

$$A_u := |\phi_2^+(u)|, \quad B_u := |\phi_3^+(u)|.$$

Because all positive pairs incident to u lie in exactly one of $\phi_1^+(u), \phi_2^+(u), \phi_3^+(u)$, we have

$$A_u + B_u \leq d(u).$$

Define

$$\begin{aligned} f_1^+ &:= \sum_{u \in U} \sum_{uv \in \phi_1^+(u)} x_{uv}, & f_2^+ &:= \sum_{u \in U} \sum_{uv \in \phi_2^+(u)} x_{uv}, & f_3^+ &:= \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} x_{uv}, \\ f_1^- &:= \sum_{u \in U} \sum_{uv \in \phi_1^-(u)} (1 - x_{uv}), & f_2^- &:= \sum_{u \in U} \sum_{uv \in \phi_2^-(u)} (1 - x_{uv}). \end{aligned}$$

Then

$$\text{Top}_k(x) = f_1^+ + f_2^+ + f_3^+ + f_1^- + f_2^-.$$

Lemma 3.

$$f_1^+ + f_1^- \leq \text{Top}_k(y_C).$$

Proof. For every positive pair, $x_{uv} \leq 1$; for every negative pair, $1 - x_{uv} \leq 1$. Hence

$$f_1^+ + f_1^- \leq \sum_{u \in U} (|\phi_1^+(u)| + |\phi_1^-(u)|) \leq \sum_{u \in U} \text{cost}_C(u) \leq \text{Top}_k(y_C). \quad \square$$

Lemma 4. If $uv \in \phi_2^+$, then

$$\text{cost}_C(u) + \text{cost}_C(v) \geq \beta M_{uv}.$$

Proof. Because $uv \in \phi_2^+$, the pair uv is positive and u, v lie in the same cluster of $\mathcal{C}$, but $uv \notin E_H$. Thus

$$|N(u) \Delta N(v)| > \beta M_{uv}.$$

Take $w \in N(u) \Delta N(v)$. If $w \in N(u) \setminus N(v)$, then uw is positive and vw is negative. Since u and v are in the same cluster, either w is with them, in which case vw is a negative disagreement, or w is outside, in which case uw is a positive disagreement. So at least one disagreement is incident to u or v . The same holds if $w \in N(v) \setminus N(u)$. Summing over $w \in N(u) \Delta N(v)$ gives

$$\text{cost}_C(u) + \text{cost}_C(v) \geq |N(u) \Delta N(v)| \geq \beta M_{uv}. \quad \square$$

Lemma 5. There exists $c_2^+ \in \mathbb{R}_{\geq 0}^V$ such that

$$f_2^+ \leq \sum_{r \in V} c_2^+(r) \text{cost}_C(r),$$

and

$$c_2^+(r) \leq \frac{2A_r}{\beta d(r)} \quad \forall r \in V, \quad \|c_2^+\|_1 \leq \frac{2}{\beta} \sum_{u \in U} \frac{A_u}{d(u)}.$$

Proof. Since $x_{uv} \leq 1$,

$$f_2^+ \leq \sum_{u \in U} \sum_{uv \in \phi_2^+(u)} 1.$$

By Lemma 4,

$$1 \leq \frac{\text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(v)}{\beta M_{uv}} \quad \text{for every } uv \in \phi_2^+.$$

Hence

$$f_2^+ \leq \sum_{u \in U} \sum_{uv \in \phi_2^+(u)} \frac{\text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(v)}{\beta M_{uv}} =: \sum_{r \in V} c_2^+(r) \text{cost}_{\mathcal{C}}(r).$$

If $r \in U$, its own-endpoint coefficient is at most

$$\sum_{rv \in \phi_2^+(r)} \frac{1}{\beta M_{rv}} \leq \frac{A_r}{\beta d(r)}.$$

Its opposite-endpoint coefficient is at most

$$\sum_{u \in U: ur \in \phi_2^+(r)} \frac{1}{\beta M_{ur}} \leq \frac{A_r}{\beta d(r)}.$$

So $c_2^+(r) \leq 2A_r/(\beta d(r))$.

For the ℓ_1 -bound,

$$\|c_2^+\|_1 = \sum_{u \in U} \sum_{uv \in \phi_2^+(u)} \frac{2}{\beta M_{uv}} \leq \frac{2}{\beta} \sum_{u \in U} \frac{A_u}{d(u)}.$$

□

Lemma 6. *There exists $c_3^+ \in \mathbb{R}_{\geq 0}^V$ such that*

$$f_3^+ \leq \sum_{r \in V} c_3^+(r) \text{cost}_{\mathcal{C}}(r),$$

and

$$c_3^+(r) \leq 2 \left(\frac{A_r B_r}{\beta d(r)^2} + \frac{A_r}{\beta d(r)} + \frac{B_r}{d(r)} \right) \quad \forall r \in V,$$

and

$$\|c_3^+\|_1 \leq 2 \sum_{u \in U} \left(\frac{A_u B_u}{\beta d(u)^2} + \frac{B_u}{\beta d(u)} + \frac{B_u}{d(u)} \right).$$

Proof. Fix $uv \in \phi_3^+$. Let $\tilde{u} \in \{u, v\}$ be an endpoint of larger d -degree, and let $\tilde{v}$ be the other endpoint. Then $M_{uv} = d(\tilde{u})$, and

$$x_{uv} = 1 - \frac{|N_H(\tilde{u}) \cap N_H(\tilde{v})|}{M_{uv}} = \frac{d(\tilde{u}) - |N_H(\tilde{u}) \cap N_H(\tilde{v})|}{M_{uv}}.$$

Now every positive neighbor of $\tilde{u}$ belongs to exactly one of $\phi_1^+(\tilde{u}), \phi_2^+(\tilde{u}), \phi_3^+(\tilde{u})$, so

$$d(\tilde{u}) = |\phi_1^+(\tilde{u})| + |\phi_2^+(\tilde{u})| + |\phi_3^+(\tilde{u})|.$$

Also, because u and v lie in the same cluster and $uv \in E_H$, every $w \in \phi_3^+(\tilde{u}) \setminus N_H(\tilde{v})$ satisfies either:

- vw is positive but removed from H , so $w \in \phi_2^+(\tilde{v})$, or
- vw is negative and w is in the same cluster as v , so vw is a disagreement and w contributes to $\text{cost}_{\mathcal{C}}(\tilde{v})$.

Therefore

$$|\phi_3^+(\tilde{u})| - |N_H(\tilde{u}) \cap N_H(\tilde{v})| \leq A_{\tilde{v}} + \text{cost}_{\mathcal{C}}(\tilde{v}).$$

Since $|\phi_1^+(\tilde{u})| \leq \text{cost}_{\mathcal{C}}(\tilde{u})$, we conclude

$$x_{uv} \leq \frac{A_u + A_v + \text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(v)}{M_{uv}}. \quad (*)$$

Summing $(*)$ over $u \in U$ and $uv \in \phi_3^+(u)$,

$$\begin{aligned} f_3^+ &\leq \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \frac{A_u + A_v + \text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(v)}{M_{uv}} \\ &= T_1 + T_2 + T_3, \end{aligned}$$

where

$$\begin{aligned} T_1 &:= \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \frac{A_u}{M_{uv}}, \\ T_2 &:= \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \frac{A_v}{M_{uv}}, \\ T_3 &:= \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \frac{\text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(v)}{M_{uv}}. \end{aligned}$$

For T_1 , expand $A_u = \sum_{uw \in \phi_2^+(u)} 1$:

$$T_1 = \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \sum_{uw \in \phi_2^+(u)} \frac{1}{M_{uv}M_{uw}}.$$

By Lemma 4,

$$1 \leq \frac{\text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(w)}{\beta M_{uw}}$$

for every $uw \in \phi_2^+(u)$, so

$$T_1 \leq \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \sum_{uw \in \phi_2^+(u)} \frac{\text{cost}_{\mathcal{C}}(u) + \text{cost}_{\mathcal{C}}(w)}{\beta M_{uv}M_{uw}}.$$

Similarly,

$$T_2 \leq \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \sum_{vw \in \phi_2^+(v)} \frac{\text{cost}_{\mathcal{C}}(v) + \text{cost}_{\mathcal{C}}(w)}{\beta M_{uv}M_{vw}}.$$

Finally T_3 already has the desired form. Define c_3^+ to be the resulting coefficient vector.

We now bound its coefficients.

Pointwise bound. Fix $r \in V$.

If r appears as the u -vertex in T_1 and T_3 , its coefficient is at most

$$\sum_{rv \in \phi_3^+(r)} \sum_{rw \in \phi_2^+(r)} \frac{1}{\beta M_{rv} M_{rw}} + \sum_{rv \in \phi_3^+(r)} \frac{1}{M_{rv}} \leq \frac{A_r B_r}{\beta d(r)^2} + \frac{B_r}{d(r)}.$$

If r appears as the v -vertex in T_2 and T_3 , the same bound holds.

If r appears as the w -vertex, then from T_1 its coefficient is at most

$$\sum_{u \in U: ur \in \phi_2^+(r)} \sum_{uv \in \phi_3^+(u)} \frac{1}{\beta M_{uv} M_{ur}} \leq \sum_{u \in U: ur \in \phi_2^+(r)} \frac{B_u}{\beta d(u) d(r)} \leq \frac{A_r}{\beta d(r)},$$

because $B_u \leq d(u)$, and there are at most A_r such u . The same estimate holds for the w -contribution from T_2 . Adding the three roles gives

$$c_3^+(r) \leq 2 \left(\frac{A_r B_r}{\beta d(r)^2} + \frac{A_r}{\beta d(r)} + \frac{B_r}{d(r)} \right).$$

ℓ_1 -bound. Dropping the costs and summing coefficients directly,

$$\begin{aligned} \|c_3^+\|_1 &\leq \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \sum_{uw \in \phi_2^+(u)} \frac{2}{\beta M_{uv} M_{uw}} \\ &\quad + \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \sum_{vw \in \phi_2^+(v)} \frac{2}{\beta M_{uv} M_{vw}} + \sum_{u \in U} \sum_{uv \in \phi_3^+(u)} \frac{2}{M_{uv}}. \end{aligned}$$

The first term is at most

$$2 \sum_{u \in U} \frac{A_u B_u}{\beta d(u)^2}.$$

For the second term, fix $u \in U$ and $uv \in \phi_3^+(u)$. Then

$$\sum_{vw \in \phi_2^+(v)} \frac{2}{\beta M_{uv} M_{vw}} \leq \frac{2}{\beta M_{uv}} \sum_{vw \in \phi_2^+(v)} \frac{1}{M_{vw}} \leq \frac{2}{\beta M_{uv}} \leq \frac{2}{\beta d(u)}.$$

Summing over $uv \in \phi_3^+(u)$ gives at most $2B_u/(\beta d(u))$. The third term is at most $2B_u/d(u)$. Summing over $u \in U$ yields the claimed ℓ_1 -bound. $\square$

Lemma 7.

$$f_2^+ + f_3^+ \leq \frac{4}{\beta} \text{Top}_k(y\mathcal{C}).$$

Proof. Let $c^+ := c_2^+ + c_3^+$. By Lemmas 5 and 6,

$$f_2^+ + f_3^+ \leq \sum_{r \in V} c^+(r) \text{cost}_{\mathcal{C}}(r).$$

For the pointwise bound, use $A_r B_r / d(r)^2 \leq B_r / d(r)$, $B_r / d(r) \leq B_r / (\beta d(r))$ (since $\beta < 1$), and $A_r + B_r \leq d(r)$:

$$\begin{aligned} c^+(r) &\leq \frac{2A_r}{\beta d(r)} + 2 \left(\frac{A_r B_r}{\beta d(r)^2} + \frac{A_r}{\beta d(r)} + \frac{B_r}{d(r)} \right) \\ &\leq \frac{2A_r}{\beta d(r)} + 2 \left(\frac{B_r}{\beta d(r)} + \frac{A_r}{\beta d(r)} + \frac{B_r}{\beta d(r)} \right) = \frac{4(A_r + B_r)}{\beta d(r)} \leq \frac{4}{\beta}. \end{aligned}$$

For the ℓ_1 -bound, use $A_u B_u / d(u)^2 \leq A_u / d(u)$ and $B_u / d(u) \leq B_u / (\beta d(u))$:

$$\begin{aligned} \|c^+\|_1 &\leq \frac{2}{\beta} \sum_{u \in U} \frac{A_u}{d(u)} + 2 \sum_{u \in U} \left(\frac{A_u B_u}{\beta d(u)^2} + \frac{B_u}{\beta d(u)} + \frac{B_u}{d(u)} \right) \\ &\leq \frac{2}{\beta} \sum_{u \in U} \frac{A_u}{d(u)} + 2 \sum_{u \in U} \left(\frac{A_u}{\beta d(u)} + \frac{B_u}{\beta d(u)} + \frac{B_u}{\beta d(u)} \right) \\ &\leq \frac{4}{\beta} \sum_{u \in U} \frac{A_u + B_u}{d(u)} \leq \frac{4}{\beta} k. \end{aligned}$$

Now apply Lemma 1 with $z = y_C$ and $\alpha = 4/\beta$. □

Lemma 8.

$$f_2^- \leq \frac{2}{1-\beta} \text{Top}_k(y_C).$$

Proof. We first note a degree-comparability fact for edges of H : if $ab \in E_H$, then

$$(1-\beta)d(b) \leq d(a) \leq \frac{1}{1-\beta}d(b). \quad (\dagger)$$

Indeed, $|N(a) \triangle N(b)| \leq \beta \max\{d(a), d(b)\}$, so

$$\min\{d(a), d(b)\} \geq |N(a) \cap N(b)| \geq (1-\beta) \max\{d(a), d(b)\}.$$

Now

$$f_2^- = \sum_{u \in U} \sum_{uv \in \phi_2^-(u)} (1 - x_{uv}) = \sum_{u \in U} \sum_{uv \in \phi_2^-(u)} \frac{|N_H(u) \cap N_H(v)|}{M_{uv}}$$

so

$$f_2^- \leq \sum_{u \in U} \sum_{uv \in \phi_2^-(u)} \sum_{w \in N_H(u) \cap N_H(v)} \frac{1}{M_{uv}}.$$

Split the triples (u, v, w) according to whether $\mathcal{C}(w) = \mathcal{C}(u)$ or not.

Case 1: $\mathcal{C}(w) = \mathcal{C}(u)$. Then $vw \in E_H \subseteq E^+$ and $\mathcal{C}(v) \neq \mathcal{C}(w)$, so vw is a positive disagreement incident to w . Thus, for fixed u, w , the number of such v is at most $\text{cost}_{\mathcal{C}}(w)$. Also, by $(\dagger)$, $d(v) \geq (1-\beta)d(w)$, hence

$$M_{uv} \geq \max\{d(u), d(v)\} \geq \max\{d(u), (1-\beta)d(w)\}.$$

Therefore the total contribution of Case 1 is at most

$$\sum_{\substack{u \in U \\ w \in N_H(u) \\ \mathcal{C}(w) = \mathcal{C}(u)}} \frac{\text{cost}_{\mathcal{C}}(w)}{\max\{d(u), (1-\beta)d(w)\}}.$$

Case 2: $\mathcal{C}(w) \neq \mathcal{C}(u)$. Fix u, w . There are at most $d_H(w) \leq d(w)$ admissible vertices v . Since $M_{uv} \geq d(u)$,

$$\sum_v \frac{1}{M_{uv}} \leq \frac{d(w)}{d(u)}.$$

But now $uw \in E_H$, so $(\dagger)$ gives $d(w) \leq d(u)/(1 - \beta)$. Hence each such pair (u, w) contributes at most $1/(1 - \beta)$. Also uw is a positive edge crossing clusters, hence a disagreement incident to u . Therefore the total contribution of Case 2 is at most

$$\sum_{u \in U} \frac{\text{cost}_{\mathcal{C}}(u)}{1 - \beta}.$$

Combining the two cases, define $c^- \in \mathbb{R}_{\geq 0}^V$ by

$$f_2^- \leq \sum_{r \in V} c^-(r) \text{cost}_{\mathcal{C}}(r).$$

The coefficient of $\text{cost}_{\mathcal{C}}(u)$ from Case 2 is at most $1/(1 - \beta)$. For a fixed w , the coefficient from Case 1 is at most

$$\sum_{\substack{u \in U \\ w \in N_H(u) \\ \mathcal{C}(w) = \mathcal{C}(u)}} \frac{1}{\max\{d(u), (1 - \beta)d(w)\}} \leq \sum_{u: w \in N_H(u)} \frac{1}{(1 - \beta)d(w)} \leq \frac{d_H(w)}{(1 - \beta)d(w)} \leq \frac{1}{1 - \beta}.$$

So

$$\|c^-\|_{\infty} \leq \frac{2}{1 - \beta}.$$

Also

$$\|c^-\|_1 \leq \sum_{\substack{u \in U \\ w \in N_H(u) \\ \mathcal{C}(w) = \mathcal{C}(u)}} \frac{1}{\max\{d(u), (1 - \beta)d(w)\}} + \sum_{u \in U} \frac{1}{1 - \beta} \leq \sum_{u \in U} \frac{d_H(u)}{d(u)} + \frac{k}{1 - \beta} \leq k + \frac{k}{1 - \beta} \leq \frac{2k}{1 - \beta}.$$

Applying Lemma 1 with $\alpha = 2/(1 - \beta)$ gives the claim. $\square$

We now combine the pieces. Let $\mathcal{C}_k^*$ be an optimal clustering for the top- k objective, and abbreviate

$$\text{OPT}_k^{\text{top}} := \text{Top}_k(y_{\mathcal{C}_k^*}).$$

Using Lemmas 3, 7, and 8,

$$\text{Top}_k(x) = f_1^+ + f_1^- + f_2^+ + f_3^+ + f_2^- \leq \left(1 + \frac{4}{\beta} + \frac{2}{1 - \beta}\right) \text{OPT}_k^{\text{top}}.$$

With $\beta = \frac{1}{2}$,

$$\text{Top}_k(x) \leq (1 + 8 + 4) \text{OPT}_k^{\text{top}} = 13 \text{OPT}_k^{\text{top}}. \quad (\ddagger)$$

Now we round x .

By the Kalhan–Makarychev–Zhou metric-rounding theorem for complete graphs, any metric $x \in [0, 1]^{V \times V}$ can be rounded to a clustering $\hat{\mathcal{C}}$ such that for every vertex u ,

$$y_{\hat{\mathcal{C}}}(u) \leq 5 \text{cost}_x(u).$$

Hypothesis audit: our input graph is complete; Lemma 2 shows x is a metric; by construction $x_{uv} \in [0, 1]$ and $x_{uu} = 0$. So the theorem applies. (papers.neurips.cc)

Therefore, for every k ,

$$\text{Top}_k(y_{\hat{\mathcal{C}}}) \leq 5 \text{Top}_k(x) \leq 65 \text{OPT}_k^{\text{top}}$$

by $(\ddagger)$. This proves the stronger top- k statement.

It remains to deduce the ℓ_p -bounds.

Lemma 9. Let $y, z \in \mathbb{R}_{\geq 0}^n$. Suppose that for every $k \in \{1, \dots, n\}$,

$$\text{Top}_k(y) \leq A \text{Top}_k(z).$$

Then for every $p \in [1, \infty]$,

$$\|y\|_p \leq A \|z\|_p.$$

Proof. The cases $p = 1$ and $p = \infty$ are immediate:

$$\|y\|_1 = \text{Top}_n(y) \leq A \text{Top}_n(z) = A \|z\|_1,$$

and

$$\|y\|_\infty = \text{Top}_1(y) \leq A \text{Top}_1(z) = A \|z\|_\infty.$$

Now let $1 < p < \infty$, and let q be the Hölder conjugate, $1/p + 1/q = 1$. Sort both vectors decreasingly:

$$y_1^\downarrow \geq \dots \geq y_n^\downarrow, \quad z_1^\downarrow \geq \dots \geq z_n^\downarrow.$$

By duality,

$$\|y\|_p = \max_{\substack{w \geq 0 \\ \|w\|_q = 1}} \sum_{i=1}^n w_i y_i^\downarrow.$$

Choose a maximizer w and rearrange its coordinates decreasingly as well; this does not decrease the scalar product with $y^\downarrow$. Thus we may assume

$$w_1 \geq \dots \geq w_n \geq 0.$$

Set $w_{n+1} := 0$ and $a_k := w_k - w_{k+1} \geq 0$. Then

$$w_i = \sum_{k=i}^n a_k,$$

so

$$\sum_{i=1}^n w_i y_i^\downarrow = \sum_{i=1}^n \sum_{k=i}^n a_k y_i^\downarrow = \sum_{k=1}^n a_k \sum_{i=1}^k y_i^\downarrow = \sum_{k=1}^n a_k \text{Top}_k(y).$$

Using the hypothesis,

$$\|y\|_p \leq A \sum_{k=1}^n a_k \text{Top}_k(z) = A \sum_{i=1}^n w_i z_i^\downarrow \leq A \|z\|_p.$$

□

Apply Lemma 9 with $y = y_{\hat{\mathcal{C}}}$, $A = 65$, and $z = y_{\mathcal{C}_p^*}$, where $\mathcal{C}_p^*$ is an ℓ_p -optimal clustering. Since

$$\text{Top}_k(y_{\hat{\mathcal{C}}}) \leq 65 \text{OPT}_k^{\text{top}} \leq 65 \text{Top}_k(y_{\mathcal{C}_p^*}) \quad \forall k,$$

we get

$$\|y_{\hat{\mathcal{C}}}\|_p \leq 65 \|y_{\mathcal{C}_p^*}\|_p = 65 \text{OPT}_p \quad \forall p \in [1, \infty].$$

This completes the proof. □

Gap to the full statement

None. In fact, the theorem proved is stronger: the same clustering is a constant-factor approximation for every top- k objective, and hence for every ℓ_p -objective.

Self red-team

Counterexample attempts.

- *All-positive clique.* Then $H = G$, $x \equiv 0$, and the one-cluster solution has zero disagreement. Every inequality above becomes equality or a trivial upper bound.
- *All-negative clique.* Then H consists only of self-loops, so $x_{uv} = 1$ for $u \neq v$. The metric-rounding theorem may output all singletons, which is optimal. The decomposition puts all nontrivial edges in ϕ_2^- , exactly the case handled by Lemma 8.
- *Star instance* (one center positively connected to all leaves; leaves pairwise negative). This is the canonical obstruction for the original 2024 correlation-metric proof on ℓ_1 . Here the pre-processing step removes the misleading positive edges with highly mismatched neighborhoods; those are charged by Lemma 4. This is precisely the case our proof was built to survive.
- *Small n .* For $n = 1, 2, 3$, all denominators are positive because self-loops ensure $d(u) \geq 1$; the top- k and ℓ_p conclusions reduce to immediate checks.

Three likely referee objections.

1. “*You added self-loops; did you change the optimization problem?*” No. Self-loops are used only inside $N(u)$, $d(u)$, and H . They never contribute to disagreements, and $x_{uu} = 0$, so every clustering cost in the original problem is unchanged.
2. “*Claim 1 hides a polyhedral fact.*” I checked it explicitly: the polytope $\{x \in [0, 1]^V : \sum x \leq k\}$ has exactly the $\{0, 1\}$ -vectors with at most k ones as its extreme points, so the convex-combination argument is valid.
3. “*There is a double-counting ambiguity in the $f_i^\pm$ -sums.*” The sums are intentionally over *vertex incidences*: an edge with both endpoints in U contributes twice, exactly as it should, because the disagreement vector charges each disagreement edge to both endpoints.

Boundary cases checked.

- $k = 1$: top-1 is the ℓ_∞ -objective.
- $k = n$: top- n is the ℓ_1 -objective.
- $p = 1, \infty$: handled separately in Lemma 9.
- $\beta = \frac{1}{2}$: all places requiring $0 < \beta < 1$ remain valid.

Invoked results

- **Kalhan–Makarychev–Zhou metric rounding on complete graphs** (T00L). *Statement used.* Let $G = (V, E^+ \cup E^-)$ be a complete signed graph, and let x be any metric on V with $x_{uv} \in [0, 1]$ and $x_{uu} = 0$. Then there is a polynomial-time clustering $\mathcal{R}(x)$ such that for every $u \in V$,

$$y_{\mathcal{R}(x)}(u) \leq 5 \left(\sum_{v:uv \in E^+} x_{uv} + \sum_{v:uv \in E^-} (1 - x_{uv}) \right).$$

This is the per-vertex guarantee extracted from the proof of Theorem 4.3 in Kalhan–Makarychev–Zhou (NeurIPS 2019), and it is restated explicitly as Theorem 7 in Cao–Li–Ye (ICALP 2025). (papers.neurips.cc)

Hypothesis-match audit.

- The theorem requires a *complete* signed graph: our input graph is complete by assumption.
- It requires x to be a *metric*: proved in Lemma 2.
- It requires $x_{uv} \in [0, 1]$ and $x_{uu} = 0$: both hold by construction and Lemma 2.

Self-confidence

3 — I checked each inequality line-by-line and the argument is structurally sound, but the proof is long and coefficient-heavy, so a hidden bookkeeping slip is still possible.